

Rustem Islamov

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EDUCATION

PhD in Computer Science University of Basel , Supervisor: Aurelien Lucchi	Oct. 2023 – Pres. Basel, Switzerland
Master of Science in Applied Mathematics Institut Polytechnique de Paris Supervisor: Olivier Fercoq , Thesis supervisor: Dan Alistarh GPA: 17.65/20, transcript of records	Sept. 2021 – Aug. 2023 Palaiseau, France
Bachelor of Science in Applied Mathematics and Physics Moscow Institute of Physics and Technology Supervisor: Peter Richtárik GPA: 4.95/5 (9.27/10), Top 3 at the department, transcript of records	Sept. 2017 – July 2021 Dolgoprudny, Russia

RESEARCH INTERESTS

Deep Learning (loss landscape of neural networks), Optimization (deep learning optimizers, differential privacy and robustness in distributed optimization, model quantization, computation- and memory-efficient methods)

INTERNSHIPS AND RESEARCH VISITS

Research Visit in the group of Prof. Eduard Gorbunov MBZUAI , Host: Eduard Gorbunov	Dec. 2025 – Feb. 2026 Abu Dhabi, UAE
Internship at Distributed Algorithms and Systems Lab IST Austria , Supervisors: Mher Safaryan , Dan Alistarh	Apr. 2023 – Oct. 2023 Klosterneuburg, Austria
Internship at Machine Learning and Optimization Lab EPFL , Supervisors: Hadrien Hendrikx , Martin Jaggi	Apr. 2022 – Aug. 2022 Lausanne, Switzerland
Internship at Optimization and Machine Learning Lab KAUST , Supervisor: Peter Richtárik	Mar. 2021 – Aug. 2021 Jul. 2020 – Dec. 2020 Thuwal, Saudi Arabia

SCHOLARSHIPS, HONORS AND AWARDS

Top Reviewer for NeurIPS 2025, ICLR 2026, ICML 2026 Free registration for the conference	2025-2026
University of Basel Travel Fund Covers travel costs at EUROPT 2025; merit-based	Aug. 2025
ICLR 2025 Financial Assistance Cover registration fee; merit-based	Apr. 2025
NeurIPS@Paris Workshop Travel Grant Covers the travel expenses during the workshop; merit-based	Dec. 2024
French Embassy Scholarship Given to students enrolled to French universities with high academic achievements	Sept. 2022 – May. 2023
PhD Track Excellence Scholarship IP Paris awards merit-based excellence scholarships for students enrolled in PhD tracks	Sept. 2021 – Mar. 2022 Sept. 2022 – Mar. 2023
Increased State Academic Scholarship Given to 4 year Bachelor and Master students at MIPT with scientific achievements	Feb. 2021 – June 2021 Sept. 2020 – Jan. 2021
Abramov scholarship Given to 1-3 year students with the best grades at MIPT	Sept. 2017 – June 2020

SELECTED PUBLICATIONS (TOTAL: 2 NEURIPS, 7 ICML, 4 ICLR, 3 AISTATS, 1 UAI, 3 TMLR)

22. [R. Islamov](#), [G. Malinovsky](#), [A. Gaponov](#), [A. Lucchi](#), [P. Richtárik](#), [E. Gorbunov](#), **Byzantine-Robust and Differentially Private Federated Optimization under Weaker Assumptions**, [In Proc. of 42nd Conference on Uncertainty in Artificial Intelligence](#), 2026.

21. [R. Islamov](#), R. Machacek, A. Lucchi, A. Silveti-Falls, E. Gorbunov, V. Cevher, **On the Role of Batch Size in Stochastic Conditional Gradient Methods**, In [Proc. of 43rd International Conference on Machine Learning](#), 2026.
20. [R. Islamov](#), M. Crawshaw, J. Cohen, R. Gower, **Non-Euclidean Gradient Descent Operates at the Edge of Stability**, In [Proc. of 43rd International Conference on Machine Learning](#), (**Oral**), 2026.
18. Foivos Alimisis*, [R. Islamov](#)*, A. Lucchi (*equal contribution, alphabetical order). **Why Do We Need Warm-up? A Theoretical Perspective**, In [Proc. of 43rd International Conference on Machine Learning](#), 2025.
17. [R. Islamov](#), [N. Ajroldi](#), A. Orvieto, A. Lucchi. **Enhancing Optimizer Stability: Momentum Adaptation of NGN Step-size**, in [Advances in Neural Information Processing Systems](#), 2025.
16. E. M. Compagnoni, [R. Islamov](#), A. Orvieto, E. Gorbunov. **On the Interaction of Noise, Compression Role, and Adaptivity under μ -Smoothness: An SDE-based Approach**, In [Proc. of 43rd International Conference on Machine Learning](#), 2025.
15. [R. Islamov](#), Y. As, I. Fatkhullin. **Safe-EF: Error Feedback for Non-smooth Constrained Optimization**, In [Proc. of 42nd International Conference on Machine Learning](#), 2025.
14. [R. Islamov](#), S. Horváth, A. Lucchi, P. Richtárik, E. Gorbunov. **Double Momentum and Error Feedback for Clipping with Fast Rates and Differential Privacy**, [arXiv preprint arXiv: 2502.11682](#), 2025.
13. E. M. Compagnoni, [R. Islamov](#), F. N. Proske, A. Lucchi. **Unbiased and Sign Compression in Distributed Learning: Comparing Noise Resilience via SDEs**, in [Proc. of the 28th International Conference on Artificial Intelligence and Statistics](#), (**Oral**), 2025.
12. E. M. Compagnoni, T. Liu, [R. Islamov](#), F. N. Proske, A. Orvieto, A. Lucchi. **Adaptive Methods through the Lens of SDEs: Theoretical Insights on the Role of Noise**, in [Proc. of the 13th International Conference on Learning Representations](#), 2024.
11. [R. Islamov](#), [N. Ajroldi](#), A. Orvieto, A. Lucchi. **Loss Landscape Characterization of Neural Networks without Over-Parametrization**, in [Advances in Neural Information Processing Systems](#), 2024.
10. [R. Islamov](#)*, Y. Gao*, S. Stich(*equal contribution). **Towards Faster Decentralized Stochastic Optimization with Communication Compression**, in [Proc. of the 13th International Conference on Learning Representations](#), 2024.
9. Y. Gao*, [R. Islamov](#)*, S. Stich (*equal contribution). **EControl: Fast Distributed Optimization with Compression and Error Control**, in [Proc. of the 12th International Conference on Learning Representations](#), 2023.
8. [R. Islamov](#), M. Safaryan, D. Alistarh. **AsGrad: A Sharp Unified Analysis of Asynchronous-SGD Algorithms**, in [Proc. of the 27th International Conference on Artificial Intelligence and Statistics](#), 2023.
7. K. Mishchenko, [R. Islamov](#), E. Gorbunov, S. Horváth. **Partially Personalized Federated Learning: Breaking the Curse of Data Heterogeneity**, [Transactions on Machine Learning Research](#), 2023.
6. S. Khirirat, E. Gorbunov, S. Horváth, [R. Islamov](#), F. Karay, P. Richtárik. **Clip21: Error Feedback for Gradient Clipping**, [arXiv preprint arXiv: 2305.18929](#), 2023.
5. M. Makarenko, E. Gasanov, [R. Islamov](#), A. Sadiev, P. Richtárik. **Adaptive Compression for Communication-Efficient Distributed Training**, [Transactions on Machine Learning Research](#), 2022.
4. [R. Islamov](#), X. Qian, S. Hanzely, M. Safaryan, P. Richtárik. **Distributed Newton-Type Methods with Communication Compression and Bernoulli Aggregation**, [Transactions on Machine Learning Research](#), 2022.
3. X. Qian, [R. Islamov](#), M. Safaryan, P. Richtárik. **Basis Matters: Better Communication-Efficient Second Order Methods for Federated Learning**, in [Proc. of the 25th International Conference on Artificial Intelligence and Statistics](#), 2022.
2. M. Safaryan, [R. Islamov](#), X. Qian, P. Richtárik. **FedNL: Making Newton-Type Methods Applicable to Federated Learning**, In [Proc. of 39th International Conference on Machine Learning](#), 2022.
1. [R. Islamov](#), X. Qian, P. Richtárik. **Distributed Second Order Methods with Fast Rates and Compressed Communication**, In [Proc. of 38th International Conference on Machine Learning](#), 2021.

REVIEWING AND TEACHING

Teaching Assistant for Continuous Optimization course	Spring Semesters 2024-2026
Reviewer for Transactions on Machine Learning Research (TMLR)	2023-2026
Reviewer for Conference on Neural Information Processing Systems	2024-2025
Reviewer for International Conference on Machine Learning (ICML)	2024-2026
Reviewer for Journal on Machine Learning Research (JMLR)	2024-2025
Reviewer for Journal of Parallel and Distributed Computing (JPDC)	2024
Reviewer for Journal of Optimization Theory and Applications (JOTA)	2023
Reviewer for Artificial Intelligence and Statistics Conference (AISTATS)	2023-2026
Reviewer for Neural Information Processing Systems (NeurIPS)	2024-2026
Reviewer for International Conference on Learning Representations (ICLR)	2024-2026

TALKS AND POSTERS

Invited Talk at EUROPT Conference, Links: paper	29 June-2 July, 2025
Invited Talk at Rising Stars in AI Symposium at KAUST workshop, Links: paper , slides	21 February, 2024
Talk at NTDS workshop, Links: paper	24 October, 2023
Invited Talk at CISPA , Links: paper	16 March, 2023
Talk at ETH AI Center Symposium for PhD fellows, Links: paper	9-10 February, 2023
Prerecorded Talk at KAUST Conference on Artificial Intelligence , Links: video , paper	28 April, 2021

TECHNICAL SKILLS

Programming Languages: Python (NumPy, Matplotlib, PyTorch, Pandas), C++, LaTeX
Mathematics: Calculus, Linear Algebra, Probability Theory, Convex Analysis, Deep Learning

LANGUAGES

Russian: Native
English: Advanced (C1)
French: Elementary (A2)

HOBBIES AND INTERESTS

Football, former member of student football team
Travelling, hiking, photo shooting